

MA1521 Calculus for Computing

Lecture Group 2

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Lesson 11: Tuesday 15 September 2026

(Week 6)

Remark. Q10.7 to Q10.9 are leftover exercises from the Friday class in Week 5.

Remark. Q11.1 to Q11.4 are on Section 13.8.

More questions on Section 13.8 will be set for Friday.



Remark. For a function $g(x, y)$, its graph is the surface S defined by the equation $z = g(x, y)$. Note that

$$\underline{z = g(x, y)} \iff \underline{g(x, y) - z = 0}.$$

Thus S can also be regarded as the level surface

$$f(x, y, z) = 0, \quad \text{where } f(x, y, z) = g(x, y) - z.$$

Q10.2.5. (Supplementary Exercise) Recall that the equation of the tangent plane to the graph $z = \underline{g(x, y)}$ at (x_0, y_0) is given by

$$z = g(x_0, y_0) + g_x(x_0, y_0)(x - x_0) + g_y(x_0, y_0)(y - y_0). \quad (*)$$

Derive (*) by regarding the graph of $g(x, y)$ as a level surface.

Solution: $z = g(x, y) \Leftrightarrow g(x, y) - z = 0$

Let $f(x, y, z) = g(x, y) - z$.

Then graph of f is same as the level surface $f(x, y, z) = 0$.



Take a point (x_0, y_0) . It corresponds to the point (x_0, y_0, z_0) on the graph, where $z_0 = g(x_0, y_0)$.

$$\nabla f = \langle f_x, f_y, f_z \rangle = \langle g_x, g_y, -1 \rangle$$

$$\nabla f(x_0, y_0, z_0) = \langle g_x(x_0, y_0), g_y(x_0, y_0), -1 \rangle.$$

So the equation of the tangent plane to S at (x_0, y_0, z_0) is:

$$\nabla f(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - g(x_0, y_0) \rangle$$

same as (4) $\Leftrightarrow g_x(x_0, y_0) \cdot (x - x_0) + g_y(x_0, y_0) \cdot (y - y_0) - (z - g(x_0, y_0)) = 0$

$\Leftrightarrow z = g(x_0, y_0) + g_x(x_0, y_0) \cdot (x - x_0) + g_y(x_0, y_0) \cdot (y - y_0).$

Q10.4. Consider the function $f(x, y) = e^{2x+y}$.

(a) Find the linearization of f at the point $(0, 0)$.

(b) Find the linearization of f at the point $(1, 2)$.

[Remark: In both (a) and (b), give your answers in the form $ax + by + c$.]

Solution: Given $f(x, y) = e^{2x+y}$. Then

$$f_x(x, y) = 2e^{2x+y}, \quad f_y(x, y) = e^{2x+y}.$$

(a) The linearization of f at $(0, 0)$ is:

$$\begin{aligned} L_1(x, y) &= f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) \\ &= e^{2 \cdot 0 + 0} + 2e^{2 \cdot 0 + 0}x + e^{2 \cdot 0 + 0}y \\ &= 1 + 2x + y \\ &= \underline{2x + y + 1}. \end{aligned}$$

(b) The linearization of f at $(1, 2)$ is:

$$\begin{aligned} L_2(x, y) &= f(1, 2) + f_x(1, 2)(x - 1) + f_y(1, 2)(y - 2) \\ &= e^{2 \cdot 1 + 2} + 2e^{2 \cdot 1 + 2}(x - 1) + e^{2 \cdot 1 + 2}(y - 2) \\ &= e^4 + 2e^4(x - 1) + e^4(y - 2) \\ &= \underline{2e^4x + e^4y - 3e^4} \end{aligned}$$

For a function $w = f(x, y)$, recall again:

$$f(x, y) \approx f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0). \quad (1)$$

Write

$$\Delta x = x - x_0, \quad \Delta y = y - y_0, \quad \Delta w = \Delta f = f(x, y) - f(x_0, y_0).$$

Then the above approximation becomes

$$\Delta w = \Delta f \approx \underline{f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y}. \quad (2)$$

With the above in mind, we introduce the differentials of the variables involved:

$$\underline{dw = df = f_x(x_0, y_0)dx + f_y(x_0, y_0)dy}. \quad (3)$$

Informally, (3) gives us a way to remember (2) as follows:
 Letting $\underline{dx} = \underline{\Delta x}$ and $\underline{dy} = \underline{\Delta y}$, then $\underline{df}$ becomes the
 LHS of (2). So (2) becomes

$$\underline{\Delta f} \approx \underline{df} = f_x(x_o, y_o) \overset{\Delta x}{dx} + f_y(x_o, y_o) \overset{\Delta y}{dy}.$$

Now if we allow (x_o, y_o) to vary, then we get

$$dw = df = f_x(x, y)dx + f_y(x, y)dy,$$

or simply

$$dw = \underline{df = f_x dx + f_y dy}.$$

Sometimes, the differential $\underline{df}$ is also called the **total differential** of f .

For a function $w = w(x, y, z)$, we similarly have

$$dw = w_x dx + w_y dy + w_z dz.$$

Q10.5. Consider the function $w = xy^2z^3$. Find the (total) differential dw in terms of dx , dy and dz . $w = w(x, y, z)$

Solution:

$$\begin{aligned} dw &= w_x dx + w_y dy + w_z dz \\ &= y^2 z^3 dx + 2xy z^3 dy + 3xy^2 z^2 dz. \end{aligned}$$

Q10.6. Consider the function $w = w(x, y, z)$ defined implicitly by the equation

$$z^2 + xyw + yw^3 = 0.$$

Find the differential dw in terms of dx , dy and dz .

[Suggestion: Regard both sides as functions of x, y, z, w and take differentials on both sides. Then solve for dw .]

[Remark. Alternatively, you may also use implicit differentiation to find w_x, w_y, w_z and then get dw .]

Solution: Given that $w = w(x, y, z)$ satisfies

$$\underline{z^2} + \underline{xyw} + \underline{yw^3} = 0.$$

Regarding both sides as functions of $x, y, z, \overset{w(x,y,z)}{w}$, and taking differentials on both sides, we get

$$\underline{2zdz} + (\underline{ywdx} + \underline{xwdy} + \underline{xydw}) + (\underline{w^3dy} + \underline{3yw^2dw}) = 0$$

$$\Rightarrow ywdx + (xw + w^3)dy + 2zdz + (xy + 3yw^2)\underline{dw} = 0$$

$$\Rightarrow dw = -\frac{ywdx + (xw + w^3)dy + 2zdz}{xy + 3yw^2}$$

$$\Rightarrow dw = \overset{w_x}{-\frac{w}{x + 3w^2}}dx - \overset{w_y}{\frac{xw + w^3}{xy + 3yw^2}}dy - \frac{2z}{xy + 3yw^2}dz.$$

Remark. The equation

$$dw = -\frac{w}{x+3w^2}dx - \frac{xw+w^3}{xy+3yw^2}dy - \frac{2z}{xy+3yw^2}dz$$

means that

$$\frac{\partial w}{\partial x} = -\frac{w}{x+3w^2}, \quad \frac{\partial w}{\partial y} = -\frac{xw+w^3}{xy+3yw^2}, \quad \frac{\partial w}{\partial z} = -\frac{2z}{xy+3yw^2}.$$

Check: Given $z^2 + xyw + yw^3 = 0$.

Taking $\partial/\partial x$ on both sides (with $w = w(x, y, z)$ in mind), we get

$$\begin{aligned} 0 + yw \cdot 1 + 3y^2 w \frac{\partial w}{\partial x} &= 0 \\ \Rightarrow \frac{\partial w}{\partial x} &= -\frac{yw}{xy+3yw^2} = -\frac{w}{x+3w^2}. \end{aligned}$$

Q10.7. Consider the function $f(x, y, z) = xz + \sin(xyz)$.

Using linear approximation, estimate the change in the value of f when (x, y, z) changes from $(\frac{1}{2}, \pi, \frac{1}{3})$ to

$(\frac{1}{2} + 0.03, \pi + 0.04, \frac{1}{3} - 0.02)$.

Solution: Given that

$$\underline{f(x, y, z)} = \underline{xz + \sin(xyz)}.$$

When (x, y, z) changes from the point $P(\frac{1}{2}, \pi, \frac{1}{3})$ to $(\frac{1}{2} + 0.03, \pi + 0.04, \frac{1}{3} - 0.02)$, we have

$$\Delta x = 0.03, \quad \Delta y = 0.04, \quad \Delta z = -0.02.$$

Then the change in the value of f is given by

$$\begin{aligned}\Delta f &\approx df = f_x dx + f_y dy + f_z dz \\ &= f_x \Delta x + f_y \Delta y + f_z \Delta z,\end{aligned}$$

where f_x, f_y, f_z are evaluated at $(\frac{1}{2}, \pi, \frac{1}{3})$.

Now $f(x, y, z) = xz + \sin(xyz)$. Thus

$$f_x = z + yz \cos(xyz),$$

$$f_y = xz \cos(xyz),$$

$$f_z = x + xy \cos(xyz).$$

So

$$f_x\left(\frac{1}{2}, \pi, \frac{1}{3}\right) = \frac{1}{3} + \pi \cdot \frac{1}{3} \cdot \cos \frac{\pi}{6} = \frac{1}{3} + \frac{\sqrt{3}}{6}\pi,$$

$$f_y\left(\frac{1}{2}, \pi, \frac{1}{3}\right) = \frac{1}{2} \cdot \frac{1}{3} \cdot \cos \frac{\pi}{6} = \frac{\sqrt{3}}{12},$$

$$f_z\left(\frac{1}{2}, \pi, \frac{1}{3}\right) = \frac{1}{2} + \frac{1}{2} \cdot \pi \cdot \cos \frac{\pi}{6} = \frac{1}{2} + \frac{\sqrt{3}}{4}\pi.$$

Hence

$$\begin{aligned}\Delta f &\approx f_x\left(\frac{1}{2}, \pi, \frac{1}{3}\right)\Delta x + f_y\left(\frac{1}{2}, \pi, \frac{1}{3}\right)\Delta y + f_z\left(\frac{1}{2}, \pi, \frac{1}{3}\right)\Delta z \\ &= \left(\frac{1}{3} + \frac{\sqrt{3}}{6}\pi\right) \cdot 0.03 + \frac{\sqrt{3}}{12} \cdot 0.04 + \left(\frac{1}{2} + \frac{\sqrt{3}}{4}\pi\right) \cdot (-0.02) \\ &\approx 0.0058.\end{aligned}$$

Q10.8. Let $f(x, y)$ be a function, and (x_o, y_o) be a point in the domain of f . Consider the quadratic polynomial (in two variables) given by

$$\begin{aligned} g(x, y) = & a + b(x - x_o) + c(y - y_o) \\ & + d(x - x_o)^2 + e(x - x_o)(y - y_o) \\ & + f(y - y_o)^2, \end{aligned}$$

where a, b, c, d, e, f are constants.

Determine a, b, c, d, e, f (in terms of data on f) if

$$\begin{aligned} g(x_o, y_o) &= f(x_o, y_o), & g_x(x_o, y_o) &= f_x(x_o, y_o) \\ g_y(x_o, y_o) &= f_y(x_o, y_o), & g_{xx}(x_o, y_o) &= f_{xx}(x_o, y_o), \\ g_{xy}(x_o, y_o) &= f_{xy}(x_o, y_o), & \text{and } g_{yy}(x_o, y_o) &= f_{yy}(x_o, y_o). \end{aligned}$$

Hence write down $g(x, y)$.

$$\text{Soln : } g(x, y) = a + b(x - x_o) + c(y - y_o) + \underline{d(x - x_o)^2} + e(x - x_o)(y - y_o) + f(y - y_o)^2.$$

$$\text{Then } g_x = b + \underline{2d(x - x_o)} + e(y - y_o),$$

$$g_y = c + \underline{e(x - x_o)} + 2f(y - y_o),$$

$$g_{xx} = \underline{2d}, \quad g_{xy} = e, \quad g_{yy} = 2f.$$

$$\text{So } g(x_o, y_o) = a \implies a = f(x_o, y_o),$$

$$g_x(x_o, y_o) = b \implies b = f_x(x_o, y_o),$$

$$g_y(x_o, y_o) = c \implies c = f_y(x_o, y_o),$$

$$f_{xx}(x_o, y_o) = g_{xx}(x_o, y_o) = 2d \implies d = \frac{1}{2} \cdot f_{xx}(x_o, y_o),$$

$$g_{xy}(x_o, y_o) = e \implies e = f_{xy}(x_o, y_o),$$

$$g_{yy}(x_o, y_o) = 2f \implies f = \frac{1}{2} \cdot f_{yy}(x_o, y_o).$$

Thus

$$\begin{aligned} g(\mathbf{x}, \mathbf{y}) = & f(\mathbf{x}_o, \mathbf{y}_o) + f_x(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o) + f_y(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{y} - \mathbf{y}_o) \\ & + \frac{1}{2} f_{xx}(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o)^2 \\ & + f_{xy}(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{x} - \mathbf{x}_o)(\mathbf{y} - \mathbf{y}_o) \\ & + \frac{1}{2} f_{yy}(\mathbf{x}_o, \mathbf{y}_o)(\mathbf{y} - \mathbf{y}_o)^2. \end{aligned}$$

[Remark: $g(\mathbf{x}, \mathbf{y})$ give a quadratic approximation of $f(\mathbf{x}, \mathbf{y})$ for $(\mathbf{x}, \mathbf{y})$ near $(\mathbf{x}_o, \mathbf{y}_o)$.

Also, $g(\mathbf{x}, \mathbf{y})$ is called the 2^{nd} -degree Taylor polynomial of f at $(\mathbf{x}_o, \mathbf{y}_o)$.]

More generally, for each positive integer n , the degree n Taylor polynomial of $f(x, y)$ at (x_o, y_o) is given by

$$T_n f(x, y) = \sum_{k=0}^n \sum_{\ell=0}^k \frac{1}{(k-\ell)! \ell!} \frac{\partial^k f}{\partial x^{k-\ell} \partial y^\ell} \Big|_{(x_o, y_o)} (x - x_o)^{k-\ell} (y - y_o)^\ell.$$

Each $T_n f(x, y)$ gives an approximation of $f(x, y)$ at points (x, y) near (x_o, y_o) .

If the error in such approximation $\rightarrow 0$ as $n \rightarrow \infty$, then we get the Taylor series representation of f given by

$$f(x, y) = \sum_{k=0}^{\infty} \sum_{\ell=0}^k \frac{1}{(k-\ell)! \ell!} \frac{\partial^k f}{\partial x^{k-\ell} \partial y^\ell} \Big|_{(x_o, y_o)} (x - x_o)^{k-\ell} (y - y_o)^\ell.$$

Q10.9. Consider the function $f(x, y) = e^{x+4y} \cos x$.

(a) Find the second degree Taylor polynomial of f at $(0, 0)$.

(b) Using part (a), find an approximate value of $f(0.2, 0.1)$.

Solution: (a) Given $f(x, y) = e^{x+4y} \cos x$.

From Q10.8 (with $x_0 = 0$, $y_0 = 0$), the 2^{nd} degree Taylor polynomial of f at $(0, 0)$ is given by

$$f(0, 0) + \underbrace{f_x(0, 0)x + f_y(0, 0)y}_{1} + \frac{1}{2}f_{xx}(0, 0)x^2 + f_{xy}(0, 0)xy + \frac{1}{2}f_{yy}(0, 0)y^2.$$

$$\text{Then } f(x, y) = e^{x+4y} \cos x \implies f(0, 0) = e^{0+4 \cdot 0} \cos 0 = 1,$$

$$f_x = e^{x+4y} \cos x - e^{x+4y} \sin x \implies \underline{f_x(0, 0) = 1},$$

$$f_y = 4e^{x+4y} \cos x \implies f_y(0, 0) = 4,$$

$$\begin{aligned} f_{xx} &= e^{x+4y} \cos x - e^{x+4y} \sin x - e^{x+4y} \sin x - e^{x+4y} \cos x \\ &= -2e^{x+4y} \sin x \implies f_{xx}(0, 0) = 0 \end{aligned}$$

$$f_{xy} = 4e^{x+4y} \cos x - 4e^{x+4y} \sin x \implies f_{xy}(0, 0) = 4,$$

$$f_{yy} = 16e^{x+4y} \cos x \implies f_{yy}(0, 0) = 16.$$

Thus the 2^{nd} degree Taylor polynomial of f at $(0, 0)$ is:

$$\begin{aligned}T_2 f(x, y) &= 1 + x + 4y + \frac{1}{2} \cdot 0x^2 + 4xy + \frac{1}{2} \cdot 16y^2 \\&= 1 + x + 4y + 4xy + 8y^2.\end{aligned}$$

(b) Putting $x = 0.2$, $y = 0.1$ in (a), we get

$$\begin{aligned}f(0.2, 0.1) &\approx T_2 f(0.2, 0.1) \\&= 1 + 0.2 + 4(0.1) + 4(0.2)(0.1) + 8(0.1)^2 \\&= 1.76.\end{aligned}$$

(Actually, $f(0.2, 0.1) = e^{0.2+4(0.1)} \cos 0.2 \approx 1.786$.)

$$T_1 f(0.2, 0.1) = 1.6$$

The method of Lagrange multipliers

Goal: To maximize/minimize the function $f(x, y, z)$ subject to the constraint

$$\underline{g(x, y, z) = c.} \quad (1)$$

(Here c is a given constant.)

If $f(x, y, z)$ attains max or min at a point $P(x, y, z)$ subject to the constraint in (1), then there exists some real number λ such that

$$\nabla f(P) = \lambda \nabla g(P).$$

constrained
critical
points

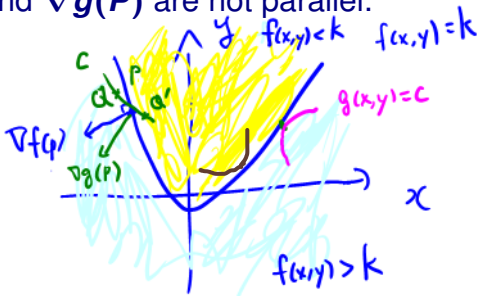
(In other words, $\nabla f(P)$ and $\nabla g(P)$ are parallel.)

Remark. Similar statements hold for functions of two variables x, y .

Graphical illustration: Consider the function in Q06.3.5:

$$\underline{f(x, y) = x^2 - y.}$$

Let $\mathbf{C}$ be the curve defined by the constraint $g(x, y) = c$. Suppose $\mathbf{P}$ is a point on $\mathbf{C}$ such that $\mathbf{f}(\mathbf{P}) = k$, but $\nabla \mathbf{f}(\mathbf{P})$ and $\nabla \mathbf{g}(\mathbf{P})$ are not parallel.



Then one sees that there are points $\mathbf{Q}, \mathbf{Q}'$ on $\mathbf{C}$ near $\mathbf{P}$ where $\mathbf{f}(\mathbf{Q}) > k$ and $\mathbf{f}(\mathbf{Q}') < k$.

Thus $f(\mathbf{x}, \mathbf{y})$ cannot attain max or min at $\mathbf{P}$ subject to the constraint $g(\mathbf{x}, \mathbf{y}) = c$.

In other words, if $f(\mathbf{x}, \mathbf{y})$ attains max or min at $\mathbf{P}$ subject to the constraint $g(\mathbf{x}, \mathbf{y}) = c$, then $\nabla f(\mathbf{P})$ and $\nabla g(\mathbf{P})$ must be parallel. So there exists some number λ such that

$$\nabla f(\mathbf{P}) = \lambda \nabla g(\mathbf{P}).$$

This leads to the **method of Lagrange multipliers** to solve the following optimization problem:

Problem: To maximize/minimize the function $f(\mathbf{x}, \mathbf{y}, \mathbf{z})$ subject to the constraint

$$g(\mathbf{x}, \mathbf{y}, \mathbf{z}) = c. \quad (1)$$

(Here c is a given constant.)

Solution/Method: Step 1: Find 'critical points' of f subject to the constraint in (1) by solving the following system of equations:

$$(*) \quad \begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z), \\ g(x, y, z) = c. \end{cases}$$

Note that the unknowns in (*) are x, y, z, λ .

Step 2: If there are additional 'boundary conditions', then

- (a) find candidate max/min points of f along the boundary.
- (b) discard those solutions in Step 1 that do not satisfy the 'boundary conditions'.

Step 3: Compute and compare the values of f at the candidate max/min points arising from Step 1 and Step 2.

Q11.1. Use the method of Lagrange multipliers to find the maximum value of the function $f(\mathbf{x}, \mathbf{y}) = \mathbf{x}\mathbf{y}$ subject to the constraint

$$(\mathbf{x} - 2)^2 + \mathbf{y}^2 = 4.$$

Solution: To maximize $f(x,y) = xy$
Subject to constraint $(x-2)^2 + y^2 = 4$

Let $g(x,y) = (x-2)^2 + y^2$ $g(x,y) = 4$.

Step: $\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} = \begin{pmatrix} y \\ x \end{pmatrix}$, $\nabla g = \begin{pmatrix} g_x \\ g_y \end{pmatrix} = \begin{pmatrix} 2(x-2) \\ 2y \end{pmatrix}$

We consider the system of equations:

Unknowns: x, y, λ $\begin{cases} \nabla f(x,y) = \lambda \nabla g(x,y) & \Leftrightarrow \begin{cases} y = \lambda \cdot 2(x-2) & \text{--- (1)} \\ x = \lambda \cdot 2y & \text{--- (2)} \\ g(x,y) = 4 & \text{--- (3)} \end{cases} \end{cases}$

(1) $\div$ (2) gives $\frac{y}{x} = \frac{\cancel{2}(x-2)}{\cancel{2}y} \Rightarrow y^2 = x^2 - 2x$ --- (4)

Subs (4) into (3), $(x-2)^2 + x^2 - 2x = 4$
 $2x^2 - 6x = 0$

$$2x^2 - 6x = 0$$

$$2x(x-3) = 0$$

$$\Rightarrow x = 0 \text{ or } x = 3$$

by (1)

Case 1: By (4), When $x = 0$, $y^2 = 0^2 - 2 \cdot 0 = 0 \Rightarrow y = 0 \Rightarrow \lambda = 0$
 $(x, y, \lambda) = (0, 0, 0)$.

Case 2: $x = 3$ by (4), $y^2 = 3^2 - 6 = 3 \Rightarrow y = \sqrt{3}, -\sqrt{3}$
 $y = \sqrt{3} \Rightarrow \lambda = \frac{\sqrt{3}}{2}$.

Case (2)(a) $(x, y, \lambda) = (3, \sqrt{3}, \frac{\sqrt{3}}{2})$

Case 2(b), $y = -\sqrt{3}$, $\lambda = -\frac{\sqrt{3}}{2}$, $(x, y, \lambda) = (3, -\sqrt{3}, -\frac{\sqrt{3}}{2})$.

Check:- The 3 solutions satisfy (1), (2), (3).

Step 3: $f(0,0) = 0 \cdot 0 = 0$

$$f(3, \sqrt{3}) = 3\sqrt{3}$$

$$f(3, -\sqrt{3}) = -3\sqrt{3}$$

f attains constrained maximum value $3\sqrt{3}$,
at the point $(3, \sqrt{3})$.